Fraud Detection in Financial Transactions

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1. Problem Statement

Credit card fraud causes billions in losses annually. Detecting fraudulent transactions in real time is critical for financial institutions. The challenge is identifying rare fraudulent patterns among millions of legitimate transactions while minimizing false positives that frustrate customers and false negatives that cause financial loss.

2. Dataset Description

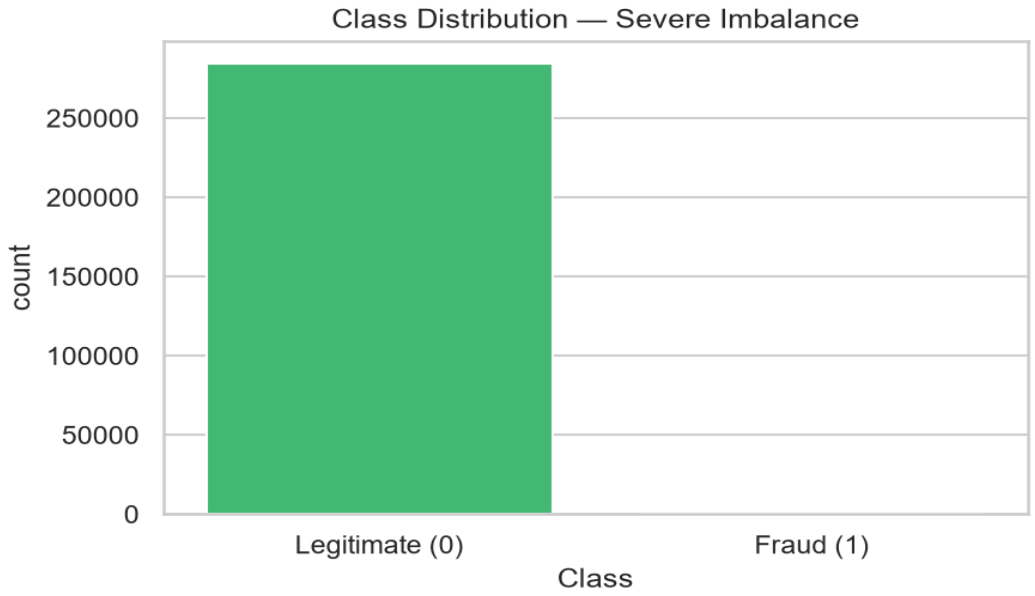
Source: Kaggle — Credit Card Fraud Detection (ULB ML Group)
URL: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud
Shape: 284,807 rows × 31 columns
Features: Time, Amount, V1–V28 (PCA-transformed), Class (target)
Legitimate transactions: 284,315
Fraudulent transactions: 492
Fraud rate: 0.173% (severe class imbalance)

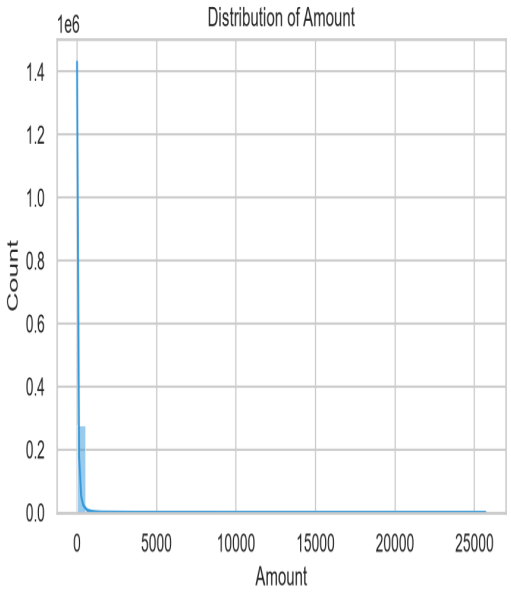
3. Methodology

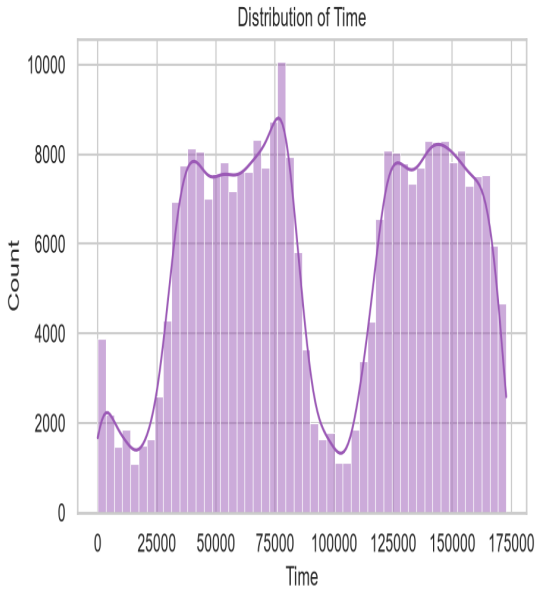
Preprocessing: Removed duplicates; scaled Amount and Time with StandardScaler (V1–V28 already PCA-normalized); 80/20 stratified train-test split.
Class imbalance: Applied SMOTE only on training data to oversample minority class.
Feature selection: Random Forest importance analysis; all 30 features retained.
Models: Logistic Regression, Decision Tree, Random Forest (100 trees), Neural Network (Dense 64→Dropout→Dense 32→Dropout→Sigmoid).
Validation: 5-fold StratifiedKFold cross-validation (ROC-AUC) on LR and RF.
Evaluation metrics: Accuracy, Precision, Recall, F1, ROC-AUC, Confusion Matrix.

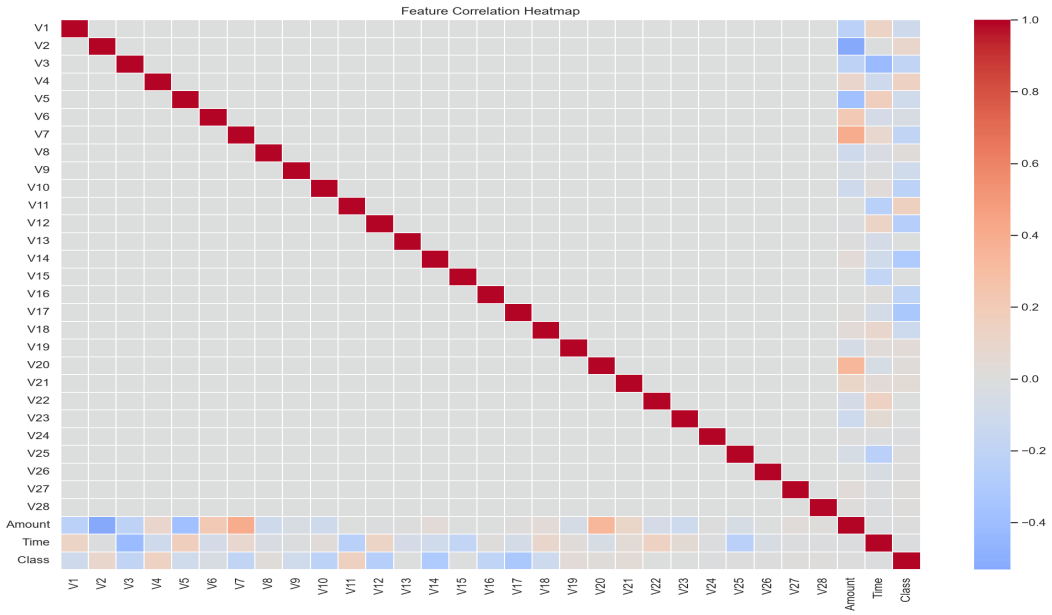
4. EDA Findings

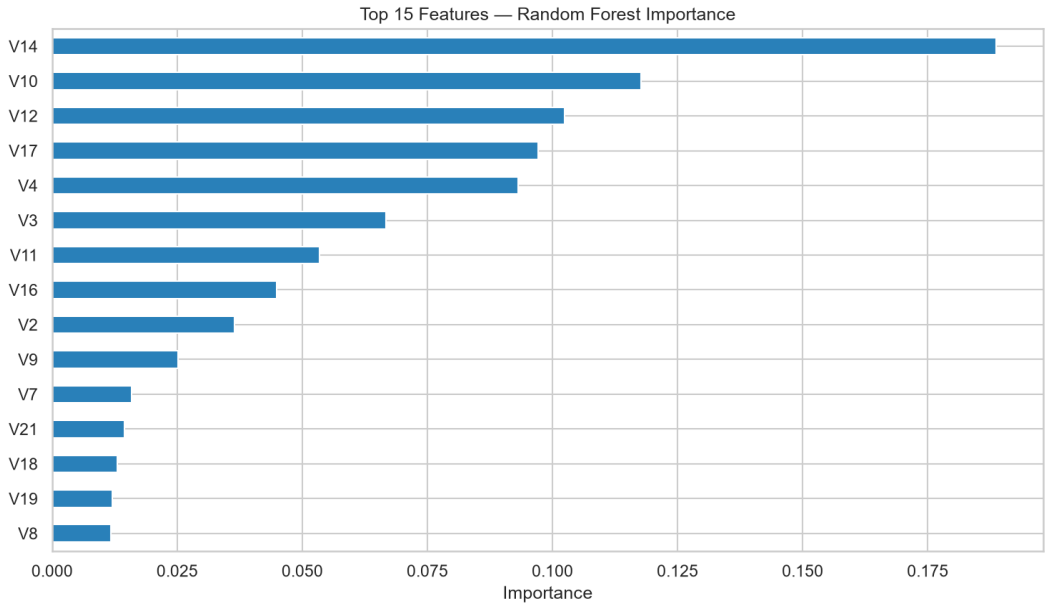
• Severe class imbalance: fraud accounts for only ~0.17% of all transactions.
• V1–V28 features (PCA components) show varying correlations with the Class label.
• Amount and Time distributions are skewed; scaling was necessary.
• Fraudulent transactions tend to differ in certain PCA components (V14, V4, V12 among top features).


Class Distribution — Severe Imbalance
count
250000
200000
150000
100000
50000
0
Legitimate (0)
Fraud (1)
Class


Distribution of Amount
1e6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Count
0
5000
10000
15000
20000
25000
Amount


Distribution of Time
10000
8000
6000
4000
2000
0
Count
0
25000
50000
75000
100000
125000
150000
175000
Time

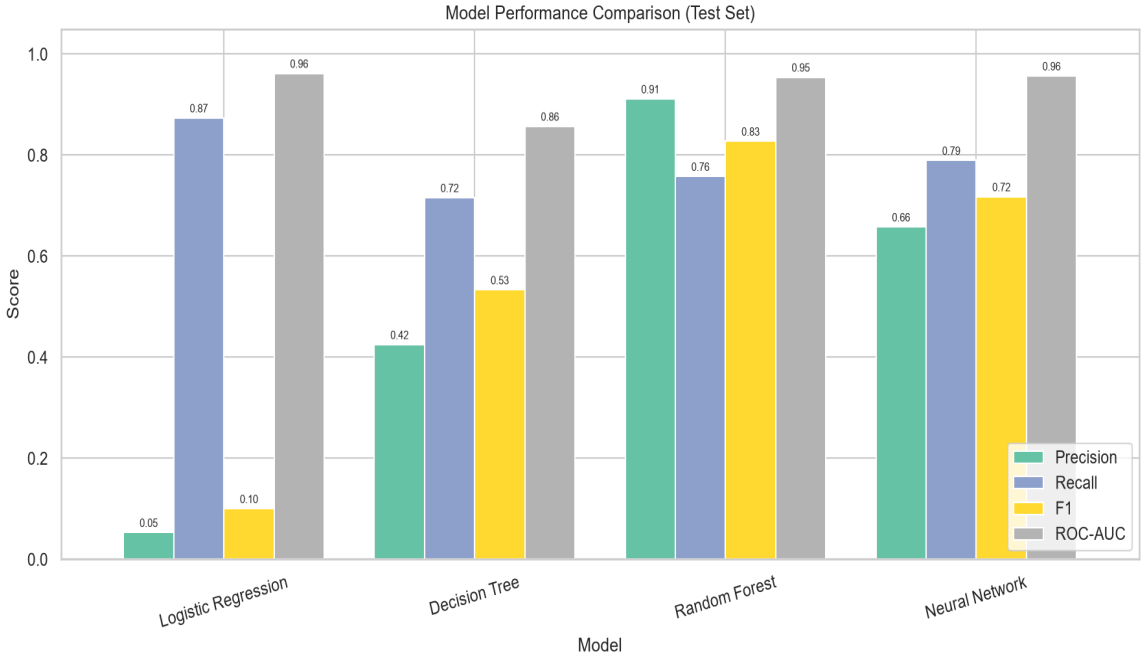

Feature Correlation Heatmap
V1
V2
V3
V4
V5
V6
V7
V8
V9
V10
V11
V12
V13
V14
V15
V16
V17
V18
V19
V20
V21
V22
V23
V24
V25
V26
V27
V28
Amount
Time
Class
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4

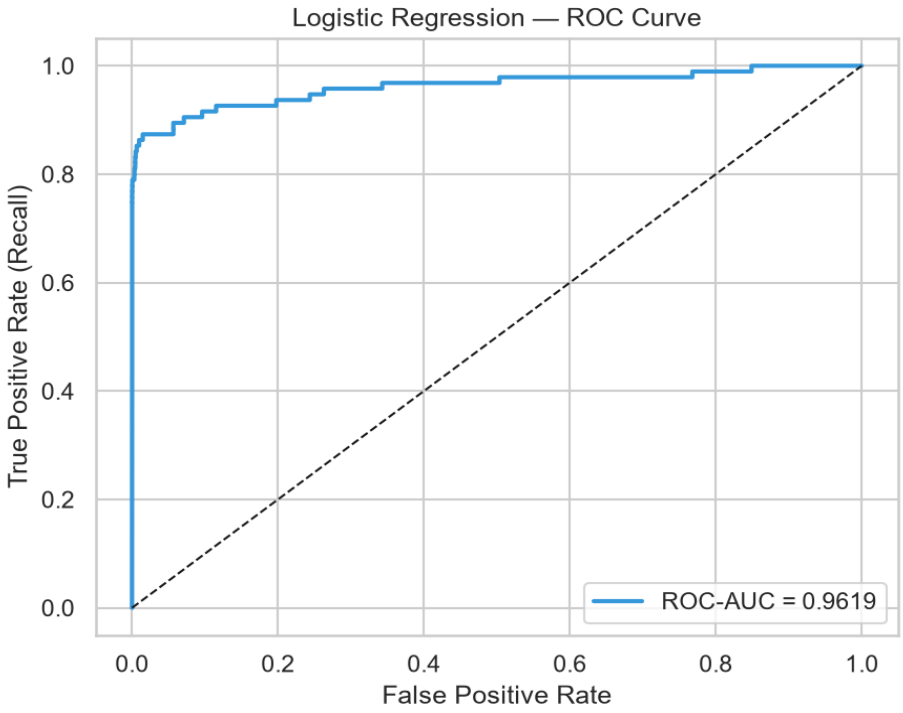

Top 15 Features — Random Forest Importance
V14
V10
V12
V17
V4
V3
V11
V16
V2
V9
V7
V21
V18
V19
V8
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Importance

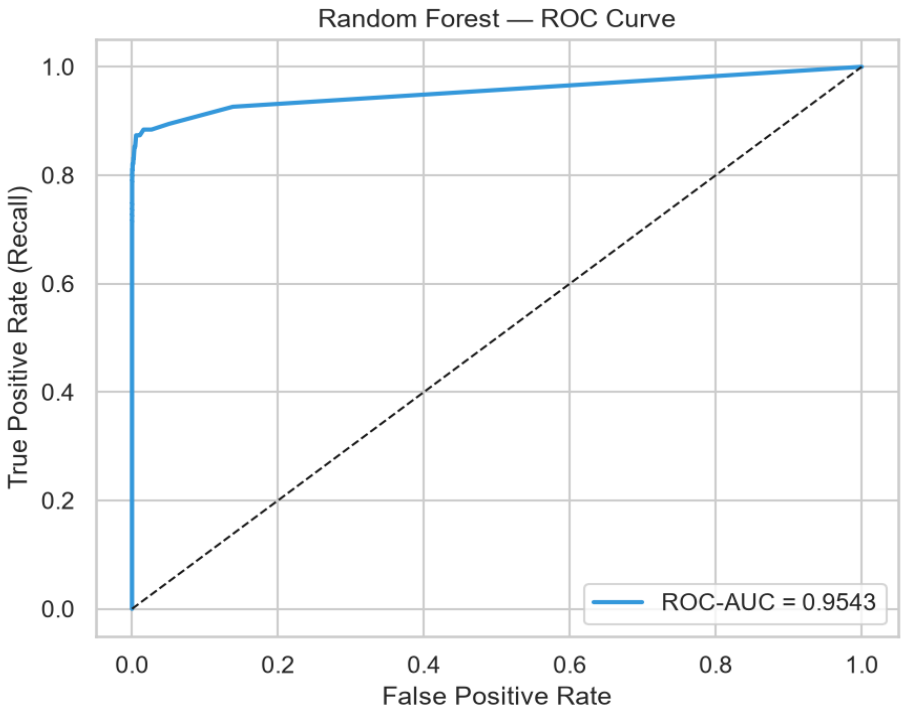
5. Model Results

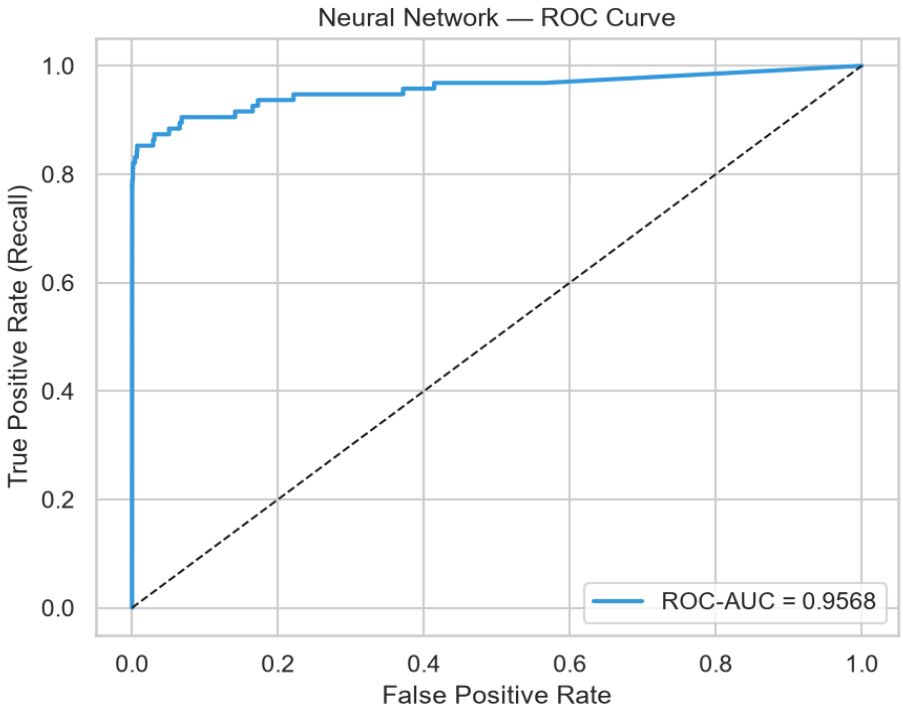
Four classifiers were trained on SMOTE-balanced training data and evaluated on the original imbalanced test set (realistic production scenario). **Recall** measures how many fraud cases we catch; **Precision** measures how many flagged transactions are actually fraud. **ROC-AUC** summarizes ranking quality.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.9737	0.0531	0.8737	0.1002	0.9619
Decision Tree	0.9979	0.4250	0.7158	0.5333	0.8571
Random Forest	0.9995	0.9114	0.7579	0.8276	0.9543
Neural Network	0.9990	0.6579	0.7895	0.7177	0.9568




Logistic Regression — ROC Curve
True Positive Rate (Recall)
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC-AUC = 0.9619


Random Forest — ROC Curve
True Positive Rate (Recall)
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC-AUC = 0.9543



Best Model (ROC-AUC): Random Forest with ROC-AUC = 0.9543.
Trade-off note: Logistic Regression achieves the highest recall (87%) and ROC-AUC, but very low precision (5%) — many false alarms. Random Forest offers a better balance (91% precision, 76% recall). The optimal choice depends on business cost of missed fraud vs. blocked legitimate transactions.

Cross-Validation Results (ROC-AUC):

Logistic Regression: Mean = 0.9807 (± 0.0079)

Random Forest: Mean = 0.9685 (± 0.0111)

6. Challenges

Class imbalance: Fraud is extremely rare; accuracy alone is misleading. SMOTE and stratified sampling were essential.
False positive tradeoff: Flagging legitimate transactions as fraud harms customer experience.
Overfitting risk: Decision Trees can memorize training data; Random Forest and regularization help.
Feature interpretability: PCA features (V1–V28) are not directly interpretable in business terms.

7. Conclusion

The Random Forest model achieved the best ROC-AUC (0.9543) on the test set. For real-world deployment, institutions should monitor recall (fraud capture rate) and precision (false alarm rate) based on business cost. Combining model scores with rule-based checks and human review provides a robust fraud prevention pipeline.

8. References

• Kaggle Dataset: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud

• GeeksforGeeks: https://www.geeksforgeeks.org/ml-credit-card-fraud-detection/

• Scikit-learn Documentation: https://scikit-learn.org/

• Dal Pozzolo et al. (2015) — Calibrating Probability with Undersampling